

Finite-Memory Fourth-Cross Support

A Weyl Certificate, Reduced Moments, and a Normalized-Memory Barrier

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Abstract

RH-107 reduced the fine-scale quotient problem to a single missing statement: the fourth projected-cross singular ratio must eventually stay above the adaptive cutoff. This paper asks whether the reduced moments of RH-95 and the finite-memory action law of RH-101 supply that statement.

Let V be an isometric packet, $P = VV^*$, and $K_t = (I - P)G_tV$, where $G_t = \sum_{j=0}^t \eta^j Q_{t-j}$ is the normalized memory Gramian. For the recent-memory truncation $\tilde{G}_{t,m}$, we prove the exact Weyl support certificate

$$\frac{s_4(K_t)}{s_1(K_t)} \geq \frac{(\hat{s}_4 - \delta_{t,m})_+}{\hat{s}_1 + \delta_{t,m}}, \quad \delta_{t,m} = \frac{\eta^m(1 - \eta^{t-m+1})}{1 - \eta}, \quad (1)$$

whenever $m \leq t$, with zero tail for a complete history. Here $\hat{s}_j$ are the singular values of $\hat{K}_{t,m} = (I - P)\tilde{G}_{t,m}V$. Equivalently, the support is certified at threshold τ when $\hat{s}_4 - \tau\hat{s}_1 \geq (1 + \tau)\delta_{t,m}$. The RH-95 identity $\hat{K}_{t,m}^* \hat{K}_{t,m} = \hat{M}_2 - \hat{A}^2$ shows that only the first two packet moments are needed for this exact finite-dimensional test; the third moment remains necessary for the full Ritz block but not for support.

On the 120 source-seeded updates at the five archived scales, a five-snapshot audit certifies every one of the 78 updates on the two finest scales for all three cutoffs $10^{-8}, 10^{-6}, 10^{-4}$. The smallest certified fine-scale ratio is 7.36045×10^{-4} , giving a minimum margin of 7.36045 at $\tau = 10^{-4}$. The moment identity is exact, but binary64 moment subtraction becomes unreliable in weak branches, with a largest observed relative discrepancy 0.7272; the support certificate therefore uses the direct recent cross action and treats moments as an algebraic reduction and consistency audit.

The result is conditional rather than all-level. We give an exact barrier family of trace-one normalized snapshots whose memory clock, packet block, complement block, and first three cross singular values are fixed while $s_4/s_1 = \varepsilon/4 \rightarrow 0$. Thus generic positivity, trace normalization, finite memory, and source seeding do not imply a positive fourth-cross lower bound. An additional physical transversality or volume law is required. No unconditional fine-support separation, Stage A theorem, Hilbert–Polya operator, zero identification, or Riemann Hypothesis result is claimed.

Keywords: projected-cross singular values; finite memory; Weyl perturbation; reduced moments; source-seeded packet; nondegeneracy barrier.

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1 Introduction

The source-seeded route of RH-94 carries a low-dimensional packet through a complete frozen prefix by repeatedly enriching it with projected-cross directions [7]. RH-95 removed the ambient cross spectral solve algebraically [6]: if $A = V^*GV$ and $M_j = V^*G^jV$, then

$$K^*K = M_2 - A^2, \quad K = (I - VV^*)GV. \quad (2)$$

RH-101 removed ambient Gram assembly from the packet action [5]. For normalized snapshot Gramians Q_t and $0 \leq \eta < 1$,

$$G_t V = \sum_{j=0}^{m-1} \eta^j Q_{t-j} V + \eta^m G_{t-m} V. \quad (3)$$

RH-107 then isolated the remaining fine-scale question [8]. For a maximum width-four selector, no quotient is proposed exactly when $s_4(K_t)/s_1(K_t) \geq \tau$.

The two earlier identities look close to closing this gate, but they solve different problems. Equation (2) is an exact measurement identity; it does not make the smallest eigenvalue of $M_2 - A^2$ positive. Equation (3) is an exact memory decomposition; its positive tail controls the error between a recent cross and the full cross, but it does not prevent the recent cross itself from becoming rank deficient.

This paper joins the two facts in the strongest elementary way. A singular value perturbation theorem turns a recent fourth-cross margin into a full memory support certificate. The certificate is useful on the two finest archived scales. A separate exact construction proves that no positive margin can follow from the generic normalized-memory data alone.

The contributions are:

1. a finite-memory Weyl certificate for the relative fourth-cross ratio;
2. a reduced-moment realization showing that support needs only M_1 and M_2 , while M_3 belongs to the downstream Ritz compression;
3. a sharpness observation for the perturbation radius;
4. an exact source-seeded normalized-memory barrier with $s_4/s_1 \rightarrow 0$ and fixed diagonal memory data;
5. a five-snapshot audit connecting the certificate to the RH-107 fine-support boundary.

2 Memory cross operators

Let $\mathcal{K} = \mathbb{R}^d$ (the complex case is identical), let $V \in \mathbb{R}^{d \times r}$ satisfy $V^* V = I_r$, and put $P = V V^*$. A normalized snapshot Gramian has the form

$$Q_j = \frac{X_j^* X_j}{\|X_j\|_F^2}, \quad Q_j \succeq 0, \quad \text{tr } Q_j = 1. \quad (4)$$

The finite-memory recursion is

$$G_{-1} = 0, \quad G_t = Q_t + \eta G_{t-1}, \quad 0 \leq \eta < 1. \quad (5)$$

The projected-cross operator at time t is

$$K_t = (I - P) G_t V. \quad (6)$$

For $m \geq 1$, define the recent operator and its cross by

$$\tilde{G}_{t,m} = \sum_{j=0}^{m-1} \eta^j Q_{t-j}, \quad (7)$$

$$\hat{K}_{t,m} = (I - P) \tilde{G}_{t,m} V. \quad (8)$$

If $m > t + 1$, the sum is understood to stop at $j = t$.

Iterating (5) gives

$$G_t - \tilde{G}_{t,m} = \eta^m G_{t-m} \quad (m \leq t), \quad (9)$$

and the difference is zero for a complete history. Since every Q_j has unit trace,

$$\text{tr } G_u = \frac{1 - \eta^{u+1}}{1 - \eta} \leq \frac{1}{1 - \eta}. \quad (10)$$

The crucial point for singular values is that the packet is an isometry: the operator norm, rather than the Frobenius norm, is enough.

3 The finite-memory support certificate

Write $\hat{s}_1 \geq \hat{s}_2 \geq \dots$ for the singular values of $\hat{K}_{t,m}$ and $s_1 \geq s_2 \geq \dots$ for those of K_t .

Theorem 3.1 (Finite-memory fourth-cross Weyl certificate). *For $m \leq t$, define*

$$\delta_{t,m} = \eta^m \frac{1 - \eta^{t-m+1}}{1 - \eta}. \quad (11)$$

Then

$$\|K_t - \hat{K}_{t,m}\|_2 \leq \delta_{t,m}, \quad (12)$$

and, whenever $\hat{s}_4 > \delta_{t,m}$,

$$\frac{s_4}{s_1} \geq \frac{\hat{s}_4 - \delta_{t,m}}{\hat{s}_1 + \delta_{t,m}}. \quad (13)$$

For $m = t + 1$, take $\delta_{t,m} = 0$ and $\hat{K}_{t,m} = K_t$.

Proof. By (9),

$$K_t - \hat{K}_{t,m} = (I - P)\eta^m G_{t-m} V.$$

Orthogonal projection and multiplication by an isometry are nonexpansive in operator norm. Since a positive semidefinite matrix is bounded in operator norm by its trace,

$$\|K_t - \hat{K}_{t,m}\|_2 \leq \eta^m \|G_{t-m}\|_2 \leq \eta^m \text{tr } G_{t-m} = \delta_{t,m}.$$

Weyl's singular-value perturbation inequality [1, 4] gives

$$s_j \geq \hat{s}_j - \delta_{t,m}, \quad s_j \leq \hat{s}_j + \delta_{t,m}.$$

Applying these statements for $j = 4$ and $j = 1$ proves (13). $\square$

Corollary 3.2 (Threshold form). *For a relative cutoff $\tau > 0$, the full cross satisfies $s_4/s_1 \geq \tau$ whenever*

$$\hat{s}_4 - \tau \hat{s}_1 \geq (1 + \tau) \delta_{t,m}. \quad (14)$$

Thus (14) is a sufficient no-quotient certificate for the maximum-width-four selector.

Proof. Rearranging (13) gives exactly (14). The adaptive support equivalence of RH-107 then implies that the selector keeps width four. $\square$

The operator-norm tail in Theorem 3.1 is sharper for support than the Frobenius packet-action estimate from RH-101. The latter contains a $\sqrt{r}$ factor because it bounds a matrix in Frobenius norm; singular-value perturbation needs only the operator norm of the positive tail.

Proposition 3.3 (Sharpness of the radius-only lower bound). *Fix $a > b > \delta \geq 0$ and set*

$$\hat{K} = \text{diag}(a, a, a, b), \quad E = -\delta e_4 e_4^*.$$

Then $\|E\|_2 = \delta$ and the fourth singular value of $\hat{K} + E$ is $b - \delta$. Hence no lower bound based only on $(\hat{s}_4, \|E\|_2)$ can improve the numerator $(\hat{s}_4 - \delta)_+$ in general.

Proof. The matrices are diagonal and their singular values are displayed directly. $\square$

4 Reduced moments and what they do not prove

For any positive semidefinite H and packet V , define

$$A_H = V^* H V, \quad M_{2,H} = V^* H^2 V, \quad K_H = (I - P) H V. \quad (15)$$

The RH-95 calculation gives the exact identity

$$C_H := K_H^* K_H = M_{2,H} - A_H^2 \succeq 0. \quad (16)$$

Consequently, for the recent memory,

$$\hat{s}_j^2 = \lambda_j \left(V^* \tilde{G}_{t,m}^2 V - (V^* \tilde{G}_{t,m} V)^2 \right). \quad (17)$$

The support certificate can therefore be evaluated in three algebraically equivalent ways:

1. form the thin action $\tilde{G}_{t,m} V$ and take the SVD of its projected cross;
2. form the small matrix C_H and diagonalize it;
3. use a validated enclosure for either of the preceding objects.

The first two are not numerically interchangeable. The difference $M_{2,H} - A_H^2$ can be much smaller than either summand. The exact identity does not prevent cancellation [2]. The third moment

$$N_H = V^* H^3 V - M_{2,H} A_H - A_H M_{2,H} + A_H^3 \quad (18)$$

is needed to assemble the complement Ritz block in RH-95, but it is not needed to decide whether the fourth cross mode is supported. This separates the current gate from the downstream Ritz-conditioning problem.

Remark 4.1 (No coercivity from the identity). Equation (16) says that the fourth mode is exactly measurable from reduced moments. It does not imply $\lambda_4(C_H) > 0$, let alone a scale-uniform lower bound for $\lambda_4(C_H)/\lambda_1(C_H)$. Such a lower bound is a separate nondegeneracy statement.

5 An exact normalized-memory barrier

We now show that generic assumptions used by RH-95 and RH-101 cannot supply the missing nondegeneracy. Let $V = [e_1, e_2, e_3, e_4]$ in $\mathbb{R}^8$ and let $U = [e_5, e_6, e_7, e_8]$. For $0 \leq \varepsilon \leq 1$, define

$$D_\varepsilon = \text{diag}(4, 3, 2, \varepsilon), \quad Q_\varepsilon = \frac{1}{34} \begin{pmatrix} I_4 & D_\varepsilon \\ D_\varepsilon & \text{diag}(16, 9, 4, 1) \end{pmatrix}. \quad (19)$$

Theorem 5.1 (Source-seeded normalized-memory barrier). *The matrix Q_ε is a trace-one positive semidefinite snapshot Gramian. Let*

$$Q_{\text{src}} = \frac{1}{12} \text{diag}(2, 2, 2, 2, 1, 1, 1, 1), \quad G_1 = Q_\varepsilon + \eta Q_{\text{src}}. \quad (20)$$

The unique leading rank-four packet of Q_{src} is V . With this source packet,

$$\text{tr } G_1 = 1 + \eta, \quad (21)$$

$$V^* G_1 V = \left(\frac{1}{34} + \frac{\eta}{6} \right) I_4, \quad (22)$$

$$U^* G_1 U = \frac{1}{34} \text{diag}(16, 9, 4, 1) + \frac{\eta}{12} I_4, \quad (23)$$

$$s(K_1) = \frac{1}{34} (4, 3, 2, \varepsilon), \quad (24)$$

$$\frac{s_4(K_1)}{s_1(K_1)} = \frac{\varepsilon}{4}. \quad (25)$$

Table 1: Certified support counts. Each entry is the number of updates whose five-snapshot lower bound clears the indicated threshold, out of the total updates at that scale (two channels combined). The last two scales are fully certified for every threshold.

threshold	0.16	0.08	0.04	0.02	0.01	fine margin
10^{-8}	4/8	11/12	20/22	34/34	44/44	7.36045
10^{-6}	1/8	10/12	20/22	34/34	44/44	736.045
10^{-4}	0/8	4/12	16/22	34/34	44/44	7.36045

In particular, the memory clock and both diagonal packet/complement blocks are independent of ε , while the fourth-cross ratio can be zero.

Proof. The block matrix in (19) is the sum of four rank-one or positive two-by-two blocks. More explicitly, define the 5×8 matrix whose rows are

$$e_1^* + 4e_5^*, \quad e_2^* + 3e_6^*, \quad e_3^* + 2e_7^*, \quad e_4^* + \varepsilon e_8^*, \quad \sqrt{1 - \varepsilon^2} e_8^*. \quad (26)$$

Its squared Frobenius norm is 34, and its normalized Gramian is exactly Q_ε . Thus $Q_\varepsilon \succeq 0$ and $\text{tr } Q_\varepsilon = 1$.

The source matrix Q_{src} has eigenvalue $2/12$ on $\text{Ran } V$ and $1/12$ on its complement, so its top rank-four packet is V . Since Q_{src} has no packet/complement cross block, the cross of G_1 is the cross of Q_ε :

$$(I - P)G_1V = UD_\varepsilon/34.$$

The remaining identities follow by taking the displayed blocks and singular values. $\square$

Remark 5.2. The construction is compatible with the finite-memory snapshot class, not with a claim about the specific folded-Gaussian production operator. If an actual one-step matrix realization is desired, take square state factors of Q_{src} and Q_ε and set $A = cQ_\varepsilon^{1/2}Q_{\text{src}}^{-1/2}$ with $c > 0$ small enough that $\rho(A) < 1$; multiplying a state by c does not change its normalized snapshot Gramian. The barrier therefore identifies missing physical input, not a failure of the finite-memory algebra.

The barrier also clarifies the role of moments. The full matrix $M_2 - A^2$ changes in its fourth direction as ε changes, so exact moments can detect the collapse. What they do not provide automatically is a lower bound on that changing eigenvalue. Any all-level proof must control that eigenvalue by a model-specific transversality, determinant, or volume estimate.

6 Five-scale certificate audit

We replay the two source-seeded channels at

$$\sigma \in \{0.16, 0.08, 0.04, 0.02, 0.01\}$$

using the RH-94 frozen horizons and the RH-96 adaptive packet updates [9]. The memory parameter is $\eta = 1/512$, the recent depth is $m = 5$, and the three relative cutoffs are $10^{-8}, 10^{-6}, 10^{-4}$. At each update we compute the recent cross action, apply Theorem 3.1, and compare the certificate with the full assembled cross. The full chain is used only as the finite comparator and to continue the inherited packet; the certificate itself uses the recent action.

The exact geometric tail bound is evaluated first. To protect the finite binary64 comparison against summation error, the audit adds an absolute 2×10^{-14} action guard. Every observed full-minus-recent cross error lies below this guarded bound. This is a finite numerical guard, not an all-level rounding theorem [3].

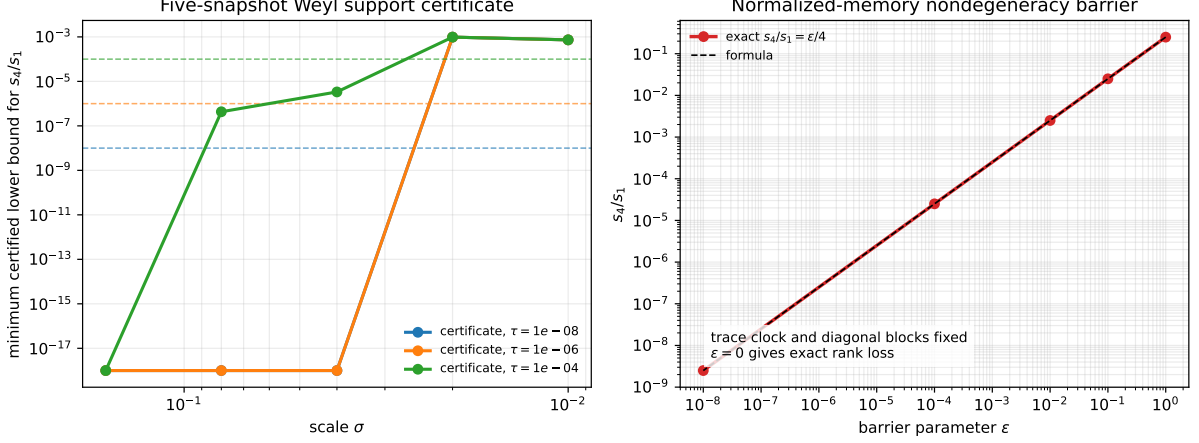


Figure 1: Left: minimum five-snapshot lower bounds at the five archived scales; dashed lines are the three selector cutoffs. Right: the exact normalized-memory barrier has fixed diagonal memory data while its fourth cross ratio follows $\epsilon/4$ to zero.

There are 360 threshold-update records. On the fine pair $\sigma \in \{0.02, 0.01\}$, all 234 threshold-update records are certified; equivalently, all 78 physical updates clear all three cutoffs. The minimum certified ratio over the fine records is

$$7.3604523 \times 10^{-4}, \quad (27)$$

which occurs at $\tau = 10^{-4}$ on the finest right channel. The maximum analytic tail bound is 2.84774×10^{-14} ; after the binary64 guard it is 4.84774×10^{-14} . No selector-equivalence failure and no observed tail-bound failure occurred.

The reduced-moment audit tells a complementary story. The largest relative discrepancy between the moment evaluation of $\hat{K}^* \hat{K}$ and the direct recent-cross Gramian is 0.7272, and the largest cancellation index is 3.89×10^{15} . These are not failures of (16); they are failures of naive binary64 subtraction in the weak regime. Direct thin actions are therefore the numerically stable implementation of the exact certificate on this dataset.

7 Route consequence and claim boundary

RH-108 closes one conditional implication that was previously implicit:

$$\begin{aligned} &\text{recent fourth-cross margin} \\ &+ \text{positive finite-memory tail bound} \end{aligned} \implies \text{full-memory no-quotient support.} \quad (28)$$

This is enough to certify the last two archived scales and explains why the observed RH-107 fine boundary is numerically robust. It does not yet make that boundary uniform in the level.

The exact barrier identifies the missing theorem in a sharper form. One must prove, for the physical source-seeded family, a lower bound of the type

$$\hat{s}_4 - \tau \hat{s}_1 \geq (1 + \tau) \delta_{t,m} + \Gamma_t, \quad \Gamma_t \geq 0, \quad (29)$$

on all sufficiently fine levels, or replace it with a model-specific volume or determinant estimate that implies the same inequality. Generic trace normalization, positivity, source packet rank, and finite-memory decay do not control Γ_t .

The claim boundary is therefore:

1. proved: the finite-memory Weyl support certificate;

2. proved: exact realization through the first two reduced moments;
3. proved: the normalized-memory/source-seed barrier;
4. validated: all fine updates at the two smallest archived scales;
5. open: an all-level physical fourth-cross lower bound;
6. open: unconditional fine-support separation and closure of the quotient supply;
7. not addressed: a Hilbert–Polya operator, zeta-zero identification, a prime-power trace formula, or the Riemann Hypothesis.

In particular, the present result moves the route forward by converting the next analytic task into a concrete nondegeneracy problem. If that problem is solved for the physical folded-Gaussian family, RH-107’s finite coarse support reduction applies. If it is not, the barrier explains exactly why the generic moment/memory architecture cannot substitute for it.

8 Reproducibility

The directory contains the certificate helpers, exact barrier constructor, full and smoke audits, figures, tests, and hash-checked archive. Run:

```
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \
  experiments/build_support_certificate_audit.py
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \
  experiments/build_support_certificate_audit.py --smoke
MPLBACKEND=Agg PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \
  experiments/make_figures.py
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/pytest -q -p no:cacheprovider
latexmk -pdf -interaction=nonstopmode -halt-on-error main.tex
cp main.pdf finite-memory-fourth-cross-support.pdf
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \
  experiments/build_archive.py
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \
  experiments/verify_archive.py
```

References

- [1] Rajendra Bhatia. *Matrix Analysis*. Springer, 1997.
- [2] Nicholas J. Higham. *Accuracy and Stability of Numerical Algorithms*. SIAM, second edition, 2002.
- [3] Siegfried M. Rump. Verification methods: Rigorous results using floating-point arithmetic. *Acta Numerica*, 19:287–449, 2010.
- [4] G. W. Stewart and Ji guang Sun. *Matrix Perturbation Theory*. Academic Press, 1990.
- [5] Liang Wang. Finite-memory packet actions for normalized gram recursions. Preprint and software repository, RH-101, 2026.
- [6] Liang Wang. Reduced projected-cross moment factorization. Preprint and software repository, RH-95, 2026.

- [7] Liang Wang. Source-seeded four-direction horizon refresh. Preprint and software repository, RH-94, 2026.
- [8] Liang Wang. Source-seeded weak-mode support laws. Preprint and software repository, RH-107, 2026.
- [9] Liang Wang. Gap-weighted weak-mode quotients. Preprint and software repository, RH-96, 2026.